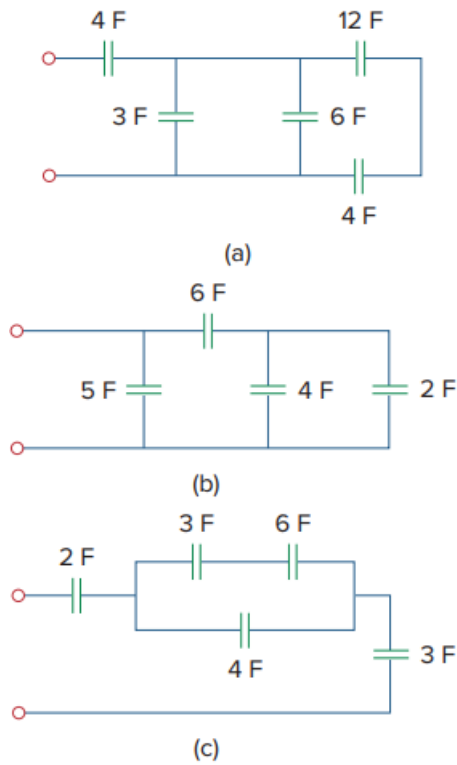


Problem

Determine the equivalent capacitance for each of the circuits of Fig. 6.51.

Fig. 6.51:



Step-by-step solution

Step 1 of 3

(A) 4 F in series with $12\text{ F} = 4 \times 12 / (16) = 3\text{ F}$

3 F in parallel with 6 F and $3\text{ F} = 3 + 6 + 3 = 12\text{ F}$

4 F in series with $12\text{ F} = 3\text{ F}$

i. e. $C_{eq} = 3\text{ F}$

Step 2 of 3

(B) $C_{eq} = 5 + [6 \times (4 + 2) / (6 + 4 + 2)] = 5 + (36 / 12) = 5 + 3 = 8\text{ F}$

Step 3 of 3

(C) 3 F in series with $6\text{ F} = (3 \times 6) / 9 = 2\text{ F}$

$$\frac{1}{C_{eq}} = \frac{1}{2} + \frac{1}{6} + \frac{1}{3} = 1$$

$C_{eq} = 1\text{ F}$